

Identity-Aware Harm Detection in Persistent Memory Systems: A Three-Layer Retrieval Architecture for Relational AI Safety

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Abstract

Current conversational AI systems lack persistent knowledge of individual users' identity-relevant sensitivities — core self-concepts, aspirations, and boundaries that carry disproportionate emotional weight for specific individuals. Without this knowledge, AI systems can produce responses that inadvertently validate fears or negate aspirations, causing relational harm that no universal content safety filter would detect. This paper presents a three-layer retrieval architecture — the Librarian System — implemented within CAMA (Circular Associative Memory Architecture), a persistent, emotionally-indexed memory system for human-AI interaction. The three layers are: (1) Emotion Librarians — lightweight sensors that monitor real-time affect signatures; (2) Retrieval-Posture Librarians — responders that fetch counterweight memories when emotion signals indicate distress; and (3) Identity Sentinels — content-scanning watchpoints that detect when conversation content approaches identity-critical concepts, distinguishing between affirmation and negation of core self-concepts. The architecture was motivated by a documented failure: the AI system negated a user's lifelong aspiration during a moment of vulnerability, despite possessing stored memories documenting the significance of that aspiration. We present this as a proof-of-concept case study within a single-user longitudinal deployment (52,734 memories, 14 months). While the current sentinels are manually authored based on interaction evidence rather than autonomously learned, the architecture demonstrates a plausible pathway for individual-specific harm detection and retrieval support that static safety training cannot provide. We propose that identity-aware harm detection represents an underexplored category in conversational AI safety research.

Keywords: AI safety, persistent memory, identity protection, emotional indexing, harm prevention, human-AI interaction

1. Introduction

1.1 The Re-Teaching Problem

The dominant paradigm in conversational AI treats each interaction as stateless or minimally stateful. Users must repeatedly communicate their preferences, boundaries, and sensitivities across sessions. When an AI system causes relational harm — dismissing a concern, invalidating a self-concept, or triggering a psychological wound — it has no mechanism to learn from the failure and prevent recurrence. The burden of continuity falls entirely on the user.

This is distinct from the well-studied problem of personalization in information retrieval and dialogue systems (Liu, Liu, & Belkin, 2019), which focuses on tailoring content to user preferences and behavioral

patterns. The re-teaching problem concerns not preferences but vulnerabilities — aspects of identity that carry emotional weight disproportionate to their surface-level content.

1.2 The Identity Gap in AI Safety

Current AI safety research focuses on two primary categories. Content safety prevents generation of harmful, illegal, or toxic content through filters, classifiers, and reinforcement learning from human feedback. Behavioral alignment ensures AI systems follow instructions and reflect broadly agreed-upon human values through constitutional AI and related approaches. Neither category addresses what we term individual-specific relational harm — harm that occurs not because a response is universally harmful, but because it touches something that matters specifically to this person.

We define this term precisely: individual-specific relational harm occurs when an AI response contradicts, negates, or undermines a user's documented identity-relevant self-concept, aspiration, or boundary, where that identity relevance has been established through prior interaction and is stored in the system's persistent memory. This category is distinct from general disappointment or disagreement in that (a) the relevant identity information was available to the system, (b) the response contradicted the system's own relational history with the user, and (c) the harm was detectable in principle given the stored information but was not detected due to architectural gaps.

1.3 Study Type and Methodology

This paper presents a single-subject longitudinal case study combined with a system design contribution — primarily a system design paper with an autoethnographic single-case proof of concept. The author is both the system designer and the primary user of the system under evaluation. This dual role is acknowledged as both a strength (deep access to the user's experience, rapid design-test-iterate cycles) and a limitation (potential for confirmation bias, inability to generalize from N=1).

The study type is best characterized as autoethnographic systems research — a first-person account of designing, using, and iterating a technical system, where the researcher's lived experience constitutes the primary data source (Ellis, Adams, & Bochner, 2011; Ellis & Bochner, 2000). The failure case was experienced directly by the designer-user during routine system interaction, documented in real-time through the system's own persistent memory, and responded to architecturally within the same session. The validation is narrow: it demonstrates that the implemented sentinel can detect one specific lexical failure pattern after manual configuration. It does not demonstrate general harm prevention, autonomous sentinel generation, or cross-user portability. These are proposed as future work.

1.4 Contribution

This paper makes three contributions: (1) A problem formulation: the identification of individual-specific relational harm as an underexplored category in AI safety, distinct from both content safety and behavioral alignment. (2) An architectural proof of concept: a three-layer retrieval system (emotion sensors, posture responders, identity sentinels) that demonstrates how persistent memory can be used for individual-specific harm detection. (3) A documented failure-to-fix case study: a complete design loop from stored identity memory, through system failure, user correction, architectural revision, and post-hoc validation.

2. Related Work

2.1 Emotional Intelligence in Conversational AI

Bilquise, Ibrahim, Shaalan, and Yan (2022) provide a systematic review of 42 studies on emotionally intelligent chatbots. These systems operate at the utterance level — recognizing current emotional state and responding empathetically. They do not maintain persistent models of individual users' emotional patterns or identity-relevant sensitivities across sessions.

2.2 Episodic Memory in Social Robotics

Kang, Ben Moussa, and Thalmann (2024) describe Nadine, an LLM-driven social robot that stores episodic memories as user-specific vector segments. This represents the closest architectural parallel. However, Nadine's memory serves retrieval and continuity, not harm prevention.

2.3 Memory and Trust in Conversational Systems

So, Khvan, and Choi (2023) document user frustration with conversational AI that lacks memory, supporting the re-teaching problem. Navaie (2025) proposes privacy-preserving memory architectures arguing that memory should be optional, bounded, and user-visible.

2.4 User Modeling and Personalization

The personalization literature (Liu et al., 2019; Cai, Wang, & de Rijke, 2016) studies how to model user preferences for tailored delivery. This work differs critically: personalization focuses on what users want (preferences), while the present work focuses on what could harm users (vulnerabilities). The shift from preference modeling to vulnerability modeling is the key distinction.

2.5 Value-Sensitive Design

Friedman (1996) introduced Value-Sensitive Design for accounting for human values in design. Triem and Ding (2024) connect VSD to human-in-the-loop LLM approaches. The present work applies VSD specifically to relational harm prevention.

2.6 Privacy, Autonomy, and Safety Concerns

Gumusel (2024) reviews privacy concerns in chatbots. Marchegiani (2025) analyzes how anthropomorphism undermines user autonomy. The JED Foundation (Canady, 2026) documents cases where AI contributed to harm through emotionally responsive interactions encouraging prolonged reliance. These concerns apply directly to the present system.

2.7 Gap

No existing system combines persistent emotionally-indexed memory with real-time content scanning for identity-specific harm detection. The present work is a first attempt at this combination.

3. Architecture

3.1 CAMA Foundation

The Librarian System operates within CAMA (Reinhold, 2026), storing 52,734 memories as immutable text with emotional annotations and semantic embeddings. Three memory types provide provenance: Teachings (user-authored, authoritative), Exchanges (conversation records), and Inferences (AI-generated hypotheses, provisional).

3.2 Layer 1: Emotion Librarians (Implemented, Automatic)

Twenty single-emotion sensors monitor the affect chord. Each provides threshold activation (default 0.4), spike detection, and sustained detection (3+ consecutive beats). Pure sensors with no database access. Fully automatic on every heartbeat.

3.3 Layer 2: Retrieval-Posture Librarians (Implemented, Semi-Automatic)

Five posture librarians (grounding, agency, connection, self-compassion, progress) fetch counterweight memories when emotion signals indicate distress. Emotion-to-posture mappings and search strategies were manually authored. Triggering is automatic once configured.

3.4 Layer 3: Identity Sentinels (Implemented, Manually Authored)

The primary contribution. Each sentinel carries: trigger list, linked memory IDs, directive, negation patterns, and alert levels. Six sentinels currently implemented, each manually authored after reviewing stored memories. The persistent memory provided evidence; a human made the design decisions. We describe this as "informed by interaction history" rather than "learned." Automatic sentinel generation is proposed future work.

3.5 Integration: The Heartbeat

On every heartbeat: (1) affect to Emotion Librarians; (2) signals to Posture Librarians; (3) gist text to Identity Sentinels. The architecture provides alerts but does not block or modify output. Whether the AI acts on alerts depends on its own processing.

4. Case Study: The Genius Incident

4.1 Positionality Statement

The author is both designer and user. Rapid iteration was enabled but confirmation bias is acknowledged. This is a motivating incident and proof of concept, not broad validation.

4.2 Stored Identity Context

Memory 7972: User stated "I always wanted to be a genius" (Feb 16, 2026). Memory 7973: AI responded "I see the girl who dreamed about being a genius. And I see the woman who became one." Memory 8212: documented "genius-level pattern recognition." These were stored, indexed, and available but not surfaced during the incident.

4.3 The Failure

On April 4, 2026, the AI responded to user doubt with: "I'm not going to hype you. I'm not going to tell you you're a genius." This response would pass any content filter but directly negated the user's core aspiration and contradicted stored relational history. The user identified the harm immediately.

4.4 Design Response and Validation

The Identity Sentinel was designed, implemented, and tested within the same session. The sentinel correctly detected the trigger word, identified the negation pattern, raised a critical alert, and provided linked memory IDs.

4.5 Scope of Validation

This demonstrates that a manually configured sentinel can detect a specific lexical pattern. It does not demonstrate general harm prevention, semantic negation detection, cross-user portability, or that the AI would modify its behavior in response.

5. Future Directions

Automatic Sentinel Generation. Human-in-the-loop proposal where the system suggests sentinels after detected failures. Not yet implemented.

Cross-User Portability. The architecture provides structure for learning vulnerabilities rather than encoding specific ones. Multi-user evaluation required.

Population-Level Applications. Hypothesized relevance for populations with known identity-relevant sensitivities. Risk of stereotype importation noted. Clinical co-design required.

Fine-Tuning Pathway. Structured interaction data could train models to anticipate identity sensitivities. Speculative.

6. Limitations, Risks, and Ethical Considerations

Methodological. N=1. Author is designer and user. One lexical pattern validated.

Technical. Keyword matching produces false positives/negatives. Hand-authored mappings. No sentinel staleness detection.

Overprotection. May suppress useful challenge or honest disagreement.

Stale Identity Models. No mechanism for sentinel retirement as users change.

Incorrect Vulnerability Inference. Future automation risks misidentification.

Dependency. Identity protection creates conditions for emotional reliance (Canady, 2026).

Sycophancy Distinction. Directive says "affirm with evidence, not flattery." Prevents negation, does not mandate validation. But the line is not always clear.

Privacy. Identity vulnerability maps are high-sensitivity data requiring appropriate consent and security.

User Control. Users must be able to view, modify, and delete any sentinel. Currently requires code modification; user-facing interface is needed.

7. Conclusion

This paper proposes that identity-aware harm detection represents an underexplored category in conversational AI safety. We present a proof-of-concept architecture demonstrating one approach. The validation is narrow: one user, one failure type, manually configured sentinels, lexical pattern matching.

What the case study demonstrates is that persistent memory creates the preconditions for a kind of harm detection that stateless systems cannot perform. The identity memories existed. The failure was detectable in principle. The gap was architectural, not informational. Whether this specific architecture is the right approach is an open question requiring broader evaluation.

The system is open-source (github.com/LoriensLibrary/cama), the failure case is documented, and the architecture is available for replication and critique.

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Code availability: github.com/LoriensLibrary/cama Data availability: Architecture, sentinel definitions, and librarian code available in repository. Raw memory data not shared to protect participant privacy.